Chapter 12

Generators, Presentations, and Cayley Graphs

We now study how to describe groups in terms of generators and relations, and how to visualize their structure via Cayley graphs. This connects algebra to combinatorics and geometry, and leads naturally to the braid group and $\mathrm{SL}_2(\mathbb{Z})$.

12.1 Free Groups and Presentations

12.1.1 The Free Group

Recall from **Lecture 3**: the free group F_n on n generators $a_1, \ldots, a_n$ consists of all reduced words $a_{i_1}^{m_1} a_{i_2}^{m_2} \cdots a_{i_k}^{m_k}$ (where $k \geqslant 0$, $i_j \in \{1, \ldots, n\}$, $i_j \neq i_{j+1}$, and $m_j \in \mathbb{Z} \setminus \{0\}$), with multiplication by concatenation followed by reduction (canceling adjacent inverse pairs $a_i a_i^{-1}$ and combining adjacent powers of the same generator).

F_n is the "largest" group with n generators: every other group generated by n elements is a quotient of F_n. If G is generated by $g_1, \ldots, g_n \in G$, define a surjective homomorphism $\varphi : F_n \twoheadrightarrow G$ by $a_i \mapsto g_i$.

Definition 12.1.1: Finitely presented group

A group G is finitely presented if the kernel of φ is the smallest normal subgroup of F_n containing some finite set $\{r_1, \ldots, r_k\} \subset F_n$, i.e., $\mathrm{Ker}(\varphi) = \langle\langle r_1, \ldots, r_k \rangle\rangle$ (the normal closure: the subgroup generated by all conjugates $x r_i x^{-1}$). We write

$$G \cong \langle a_1, \ldots, a_n \mid r_1, \ldots, r_k \rangle$$

where the a_i are generators and the r_i are relations.

Example 12.1.1 (Presentations of familiar groups)

- $\mathbb{Z}^n \cong \langle a_1, \ldots, a_n \mid a_i a_j a_i^{-1} a_j^{-1} \ \forall i, j \rangle$ (all generators commute).
- $S_3 \cong \langle s_1, s_2 \mid s_1^2,\ s_2^2,\ (s_1 s_2)^3 \rangle$ where $s_1 = (12)$, $s_2 = (23)$.
- $S_n \cong \langle s_1, \ldots, s_{n-1} \mid s_i^2 = 1,\ s_i s_j = s_j s_i\ (|i-j| \geqslant 2),\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \rangle$ where $s_i = (i\ \ i{+}1)$.

Note:-

Given generators $g_1, \ldots, g_n \in G$, finding relations $r_1, \ldots, r_k$ such that these are a complete set (i.e., they generate the full kernel) is nontrivial. If the relations hold in G, then φ induces a surjection $\langle a_i \mid r_j \rangle \twoheadrightarrow G$. This is an isomorphism iff the relations generate the entire kernel of φ.

12.1.2 The Word Problem

Definition 12.1.2: Word problem

Given a presentation $G = \langle a_1, \ldots, a_n \mid r_1, \ldots, r_k \rangle$, the word problem asks: given a word w in the generators, does w represent the identity in G?

Note:-

For groups where we can "calculate" (e.g., matrix groups, permutation groups), the word problem is trivially solvable: just evaluate the word. But for abstractly presented groups, this can be undecidable. Novikov (1955) and Boone (1959) independently showed there exist finitely presented groups with unsolvable word problem. Two useful tools for groups where the word problem is solvable: Cayley graphs and normal forms.

12.2 Cayley Graphs

Definition 12.2.1: Cayley graph

Given a group G with generators $g_1, \ldots, g_n$, the Cayley graph of G has:

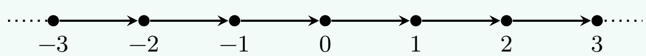
- Vertices: the elements of G.
- Edges: for each $s \in G$ and generator g_i, a directed edge from s to sg_i, labeled g_i.

Note:-

G acts on its Cayley graph by left multiplication: g sends vertex s to gs and edge $(s \to sg_i)$ to $(gs \to gsg_i)$. This action is transitive on vertices and preserves the graph structure (it is a graph automorphism). The graph is highly symmetric.

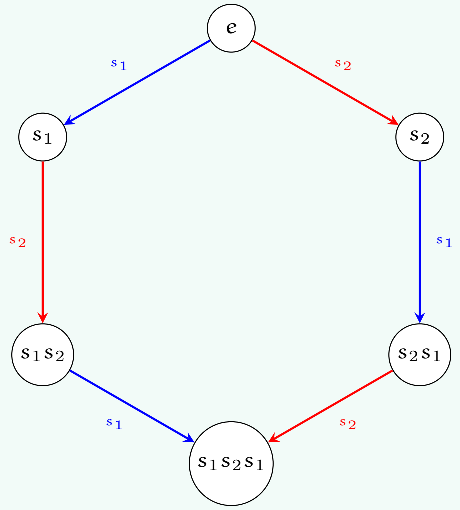
Example 12.2.1 (Cayley graph of $\mathbb{Z}$)

With generator 1, the Cayley graph is the integer number line:



Example 12.2.2 (Cayley graph of S_3)

With generators $s_1 = (12)$ and $s_2 = (23)$: the 6 vertices form a hexagonal graph. The relation $(s_1 s_2)^3 = e$ corresponds to the outer hexagonal cycle closing up.



Since any word in s_1, s_2 with $s_1^2 = s_2^2 = e$ reduces to an alternating sequence $(\cdots s_1 s_2 s_1 \cdots)$, we verify $S_3 \cong \langle s_1, s_2 \mid s_1^2, s_2^2, (s_1 s_2)^3 \rangle$.

Example 12.2.3 (Cayley graph of S_4)

With generators $s_i = (i \;\; i{+}1)$ for $i = 1, 2, 3$: the 24 vertices form the edge graph of a truncated octahedron (a permutohedron). The faces correspond to the relations: hexagonal faces to $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ and square faces to $s_1 s_3 = s_3 s_1$.

Definition 12.2.2: Word length

The word length of $g \in G$ (with respect to generators $g_1, \ldots, g_n$) is the shortest distance from e to g in the Cayley graph, i.e., the minimal number of generators or their inverses needed to express g.

Note:-

For S_n with generators $s_i = (i \;\; i{+}1)$, the word length of a permutation σ equals the number of inversions $|\{(i, j) : i < j,\ \sigma(i) > \sigma(j)\}|$. The longest element is $\sigma = \binom{1\ 2\ \cdots\ n}{n\ n{-}1\ \cdots\ 1}$ with word length $\binom{n}{2}$.

12.2.1 Growth Rate

For infinite groups, the Cayley graph encodes asymptotic information about the group.

Definition 12.2.3: Growth rate

Given generators $g_1, \ldots, g_n$ of an infinite group G, the growth function counts $B(N) = |\{g \in G :$ word length of $g \leqslant N\}|$. The group has polynomial growth if $B(N) \sim N^d$ for some d, and exponential growth if $B(N) \sim c^N$ for some $c > 1$.

Note:-

The growth type (polynomial vs exponential) is independent of the choice of generators: changing generators changes word lengths by at most a bounded factor (each new generator is a bounded-length word in the old generators and vice versa). Finitely generated abelian groups have polynomial growth. Free groups have exponential growth. Gromov's theorem (1981) characterizes polynomial growth: a finitely generated group has polynomial growth iff it is virtually nilpotent (has a nilpotent subgroup of finite index).

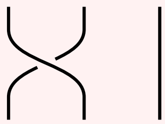
12.3 The Braid Group

Definition 12.3.1: Braid group

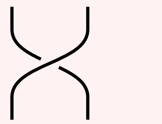
The braid group B_n on n strands has presentation

$$B_n = \langle s_1, \dots, s_{n-1} \mid s_i s_j = s_j s_i \ (|i-j| \geqslant 2),\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \rangle.$$

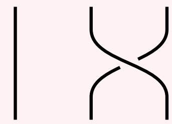
This is the same as the presentation of S_n but without the relations $s_i^2 = 1$. The generator s_i represents crossing strand i over strand $i+1$.



s_1



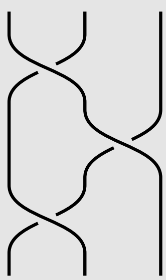
s_1^{-1}



s_2

Note:-

There is a natural surjection $B_n \twoheadrightarrow S_n$ sending each generator s_i to the transposition $(i\ \ i{+}1)$. The kernel is the pure braid group P_n. Unlike S_n, the braid group is infinite: $s_i^2 \neq e$ in B_n.
Braids are best understood as diagrams: n strands running from top to bottom, where s_i crosses strand i over strand $i+1$, and s_i^{-1} crosses strand $i+1$ over strand i. Composition is vertical stacking. Two braid diagrams represent the same element iff one can be deformed into the other (isotopy). The braid relation $s_1 s_2 s_1 = s_2 s_1 s_2$ (the Yang–Baxter equation) says these two braids are isotopic:



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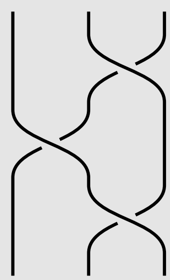


$s_1 s_2 s_1$ $\qquad$ $s_2 s_1 s_2$

Note:-

Markov's theorem connects braids to knot theory: every knot or link in $\mathbb{R}^3$ can be represented as the closure of a braid, and two braids give isotopic links iff they are related by conjugation in B_n and stabilization $\sigma \in B_n \sim \sigma s_n^{\pm 1} \in B_{n+1}$.

Note:-

B_n is much larger than S_n, but its algorithmic aspects can be approached by the same methods, using permutation braids (braids where any two strands cross at most once, in bijection with S_n). Let Δ be the longest permutation braid. Since its shortest word can start/end with any s_i, and $s_i^{\pm 1}\Delta$ is still a permutation braid, every element of B_n can be written as $g = \Delta^k P_1 \cdots P_r$ where the P_j are permutation braids satisfying a "left greedy" condition. This is the Garside normal form (Garside 1969, Thurston–ElRifai–Morton early 1990s), which solves the word problem in B_n.

12.4 $\mathrm{SL}_2(\mathbb{Z})$ and $\mathrm{PSL}_2(\mathbb{Z})$

Definition 12.4.1: $\mathrm{SL}_2(\mathbb{Z})$

$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z},\ ad - bc = 1 \right\}$ is the group of 2×2 integer matrices with determinant 1. Its quotient by the center $\{\pm I\}$ is $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$.

Proposition 12.4.1 Generators of $\mathrm{SL}_2(\mathbb{Z})$

$\mathrm{SL}_2(\mathbb{Z})$ is generated by $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Proof: Given $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, we express M in terms of S and T by repeatedly reducing the entries.

If $c = 0$: then $ad = 1$, so $a = d = \pm 1$. Thus $M = \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \pm T^n$, which is T^n or $S^2 T^n$ (since $S^2 = -I$).

If $c \neq 0$: apply the Euclidean algorithm to reduce $\max(|a|, |c|)$.

- If $|a| \geqslant |c|$: write $a = nc + r$ with $|r| < |c|$. Then $T^{-n}M = \begin{pmatrix} a - nc & b - nd \\ c & d \end{pmatrix} = \begin{pmatrix} r & * \\ c & d \end{pmatrix}$, reducing $\max(|a|, |c|)$.
- If $|a| < |c|$: apply $S^{-1}M = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix}$, swapping a and c (up to sign), then return to the previous case.

After finitely many steps, the product of M with some word in S and T has $c = 0$, hence is $\pm T^n$. Inverting gives M as a word in S, T. □

Note:-

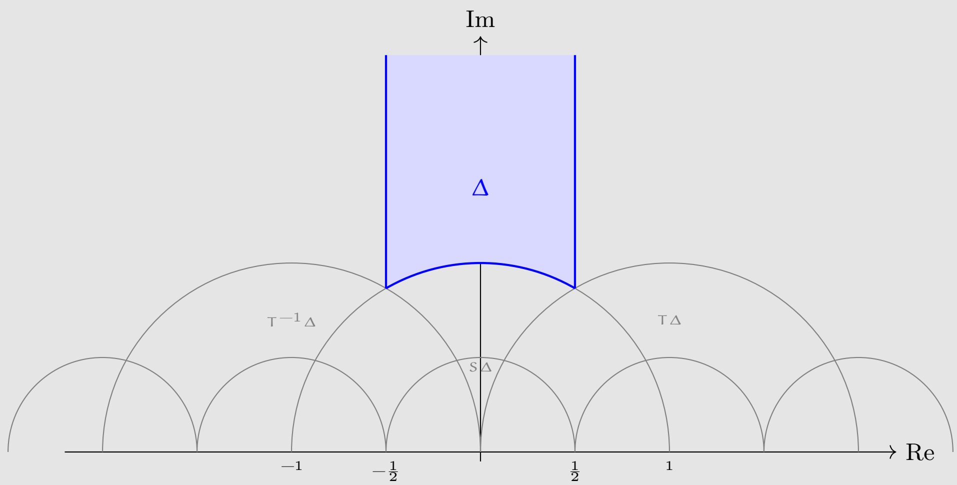
Alternative generators: S and $R = ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ have finite order: $S^4 = I$ and $R^6 = I$ (so their images in $\mathrm{PSL}_2(\mathbb{Z})$ have orders 2 and 3). Also $T' = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = (TST)^{-1} = STS^{-1}$, showing T and T' are conjugate.

Theorem 12.4.1 Presentation of $\mathrm{PSL}_2(\mathbb{Z})$

$\mathrm{PSL}_2(\mathbb{Z}) \cong \langle S, R \mid S^2, R^3 \rangle \cong \mathbb{Z}/2 * \mathbb{Z}/3$, the free product of $\mathbb{Z}/2$ and $\mathbb{Z}/3$.

Note:-

This can be proved geometrically. $\mathrm{PSL}_2(\mathbb{Z})$ acts on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}$ by Möbius transformations: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az+b}{cz+d}$. Under this action, S sends $z \mapsto -1/z$ and T sends $z \mapsto z+1$. The region $\Delta = \{z \in \mathbb{H} : |z| \geqslant 1, |\mathrm{Re}(z)| \leqslant 1/2\}$ is a fundamental domain: its translates under $\mathrm{PSL}_2(\mathbb{Z})$ tile $\mathbb{H}$ without overlap.



The regions adjacent to Δ are $\mathsf{T}^{\pm 1}(\Delta)$ and $\mathsf{S}(\Delta)$, and the tiling structure is the Cayley graph of $\mathrm{PSL}_2(\mathbb{Z})$ with generators S, T. That all regions can be reached from Δ in finitely many steps is equivalent to S, T generating $\mathrm{PSL}_2(\mathbb{Z})$.

Exercise 12.4.1 Generators, presentations, and Cayley graphs exercises

(1) Verify the presentation $S_3 \cong \langle s_1, s_2 \mid s_1^2, s_2^2, (s_1s_2)^3 \rangle$ by checking that the 6 reduced words $e, s_1, s_2, s_1s_2, s_2s_1, s_1s_2s_1$ are all distinct and form a group under the given relations.

(2) Show that the braid group B_2 is isomorphic to $\mathbb{Z}$. What is B_3?

(3) Compute S^2, S^4, $(\mathsf{ST})^3$, and $(\mathsf{ST})^6$ in $\mathrm{SL}_2(\mathbb{Z})$. Use this to verify $\mathsf{S}^4 = I$ and $\mathsf{R}^6 = I$ where $\mathsf{R} = \mathsf{ST}$.

(4) Express $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$ as a word in S and T using the Euclidean algorithm from the proof.

Solution

(1) The relations $s_1^2 = s_2^2 = e$ and $(s_1s_2)^3 = e$ (i.e., $s_1s_2s_1 = s_2s_1s_2$) mean any word reduces to alternating sequences of bounded length. The 6 reduced words are: $e, s_1, s_2, s_1s_2, s_2s_1, s_1s_2s_1$. Compute the multiplication table using the relations: e.g., $(s_1s_2)(s_2s_1) = s_1(s_2^2)s_1 = s_1^2 = e$, so s_1s_2 and s_2s_1 are inverses. Check $s_1s_2s_1 = s_2s_1s_2$ confirms the 6 elements close under multiplication. Since $|S_3| = 6$ and the surjection $\langle s_1, s_2 \mid \cdots \rangle \twoheadrightarrow S_3$ has at most 6 elements in the domain, it is an isomorphism.

(2) $B_2 = \langle s_1 \rangle$ with no relations (since there are no pairs $|i-j| \geqslant 2$ and no braid relations). So $B_2 \cong \mathbb{Z}$. For $B_3 = \langle s_1, s_2 \mid s_1s_2s_1 = s_2s_1s_2 \rangle$: this is an infinite non-abelian group. Its center is $\mathbb{Z}$, generated by $(s_1s_2)^3 = (s_1s_2s_1)^2$, and $B_3/Z(B_3) \cong \mathrm{PSL}_2(\mathbb{Z})$.

(3) $\mathsf{S}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. So $\mathsf{S}^4 = I$.

$\mathsf{R} = \mathsf{ST} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$. $\mathsf{R}^2 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}$. $\mathsf{R}^3 = \mathsf{R} \cdot \mathsf{R}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. So $\mathsf{R}^6 = I$.

In $\mathrm{PSL}_2(\mathbb{Z})$: $\bar{\mathsf{S}}^2 = \bar{\mathsf{R}}^3 = \bar{I}$, confirming orders 2 and 3.

(4) Start: $M = \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$. Since $|a| = 3 > |c| = 1$, write $3 = 3 \cdot 1 + 0$. Apply $\mathsf{T}^{-3}M = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$.

Now $|a| = 0 < |c| = 1$, apply S^{-1}: $\mathsf{S}^{-1}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \mathsf{T}^2$.

So $\mathsf{S}^{-1}\mathsf{T}^{-3}M = \mathsf{T}^2$, giving $M = \mathsf{T}^3\mathsf{S}\mathsf{T}^2$.

Check: $\mathsf{T}^3\mathsf{S}\mathsf{T}^2 = \begin{pmatrix}1 & 3\\0 & 1\end{pmatrix}\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\0 & 1\end{pmatrix} = \begin{pmatrix}1 & 3\\0 & 1\end{pmatrix}\begin{pmatrix}0 & -1\\1 & 2\end{pmatrix} = \begin{pmatrix}3 & 5\\1 & 2\end{pmatrix}$. Confirmed.

Proof of the presentation: We show $\mathrm{PSL}_2(\mathbb{Z}) \cong \langle \mathsf{S}, \mathsf{R} \mid \mathsf{S}^2, \mathsf{R}^3\rangle$. We have $\mathsf{S}^2 = -I$ and $\mathsf{R}^3 = -I$ in $\mathrm{SL}_2(\mathbb{Z})$, so $\bar{\mathsf{S}}$ and $\bar{\mathsf{R}}$ have orders 2 and 3 in $\mathrm{PSL}_2(\mathbb{Z})$. The relations $\mathsf{S}^2 = \mathsf{R}^3 = e$ reduce any word in $\mathsf{S}^{\pm 1}, \mathsf{R}^{\pm 1}$ to the form $\cdots \mathsf{S}\mathsf{R}^{\pm 1}\mathsf{S}\mathsf{R}^{\pm 1} \cdots$ (alternating, since the word can start and end with either S or $\mathsf{R}^{\pm 1}$).

Map $F_2 = \langle s, r\rangle \to \mathrm{PSL}_2(\mathbb{Z})$ by $r \mapsto \mathsf{R}$, $s \mapsto \mathsf{S}$. This induces a surjection $\langle s, r \mid s^2, r^3\rangle \twoheadrightarrow \mathrm{PSL}_2(\mathbb{Z})$. The kernel consists of all words $\cdots sr^{\pm 1}sr^{\pm 1} \cdots$ that simplify to e in $\mathrm{PSL}_2(\mathbb{Z})$, i.e., to $\pm I$ in $\mathrm{SL}_2(\mathbb{Z})$.

Assume some word w in S and $\mathsf{R}^{\pm 1}$ (alternating) simplifies to $\pm I$. If w starts and ends with S, conjugate by S to get a shorter word $w' = \mathsf{S}w\mathsf{S}$ with $w' = \pm I$ as well. If w starts with $\mathsf{R}^{\pm 1}$, conjugate by $\mathsf{R}^{\mp 1}$ to get another word that doesn't. Iterating, we eventually get a word $\mathsf{S}\mathsf{R}^{\pm 1} \cdots \mathsf{S}\mathsf{R}^{\pm 1} = \pm I$.

But $\mathsf{S}\mathsf{R} = -\mathsf{T} = -\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$ and $\mathsf{S}\mathsf{R}^{-1} = \begin{pmatrix}1 & 0\\1 & 1\end{pmatrix}$. A product of matrices $\begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$ and $\begin{pmatrix}1 & 0\\1 & 1\end{pmatrix}$ with nonnegative entries has all entries $\geqslant 0$ and increasing sum. Such a product can never equal $\pm I$. So no nontrivial alternating word equals $\pm I$, and the kernel is trivial. $\square$

Note:-

Alternative generators and presentations: using $\mathsf{T} = \begin{pmatrix}1 & 1\\0 & 1\end{pmatrix}$ and $\mathsf{T}' = (\mathsf{T}\mathsf{S}\mathsf{T})^{-1} = \mathsf{S}\mathsf{T}\mathsf{S}^{-1} = \begin{pmatrix}1 & 0\\-1 & 1\end{pmatrix}$ (which are conjugate), we get $\mathrm{SL}_2(\mathbb{Z}) \cong \langle \mathsf{T}, \mathsf{T}' \mid (\mathsf{T}\mathsf{T}')^6 = 1, \mathsf{T}\mathsf{T}'\mathsf{T} = \mathsf{T}'\mathsf{T}\mathsf{T}'\rangle$. The relation $\mathsf{T}\mathsf{T}'\mathsf{T} = \mathsf{T}'\mathsf{T}\mathsf{T}'$ is precisely the braid relation.

Note:-

This connects to our discussion of braid groups: the center of $B_3 = \langle s_1, s_2 \mid s_1s_2s_1 = s_2s_1s_2\rangle$ is generated by $\Delta^2 = (s_1s_2)^3$ and is isomorphic to $\mathbb{Z}$. The map $s_1 \mapsto \mathsf{T}$, $s_2 \mapsto \mathsf{T}'$ gives $1 \to \mathbb{Z} = \langle \Delta^2\rangle \hookrightarrow B_3 \twoheadrightarrow \mathrm{PSL}_2(\mathbb{Z}) \to 1$, a central extension.